

Infosys Springboard Virtual Internship 6.0 Completion Report

Team Details

Batch Number:13

Start date: 2/2/26

Name: K.Samyutha

Internship Duration: 8 Weeks

1. Project Title

Stock Trend AI Stock Market Forecasting Model

2. Project Objective

- The objective of this project is to develop an intelligent stock market forecasting system using deep learning techniques.
- It analyzes historical Apple Inc. stock data and applies an LSTM neural network to identify patterns in time-series data.
- The system helps investors and analysts understand potential stock price trends and supports better financial decision-making

3. Project description in detail

- ❖ The **StockTrendAI** project is a machine learning system designed to analyze historical stock market data and predict future price trends.

- ❖ It focuses on Apple Inc. (AAPL) stock data collected from the Yahoo Finance API, including attributes such as Open, High, Low, Close, and Volume.
- ❖ The dataset undergoes preprocessing steps such as removing duplicates, handling missing values, and detecting outliers using IQR and Z-score methods.
- ❖ Feature scaling using Min-Max normalization is applied, and time-series sequences are created using the previous 60 days of stock prices.
- ❖ The system evaluates performance using RMSE and MAE metrics and presents predictions through an interactive dashboard for visualization.

4. Timeline Overview

Week	Activities Planned	Activities Completed
Week 1	Understand project objectives	Studied concepts and finalized topic
Week 2	Collect stock market dataset	Retrieved and explored dataset
Week 3	Perform data preprocessing	Cleaned dataset and removed duplicates
Week 4	Apply normalization techniques	Scaled data and created sequences
Week 5	Develop LSTM prediction model	Built and configured LSTM model

Week 6	Handle missing values	Applied imputation and outlier detection
Week 7	Train and evaluate model	Calculated RMSE, MAE metrics
Week 8	Develop interactive dashboard	Built dashboard and final documentation

5a. Key Milestones

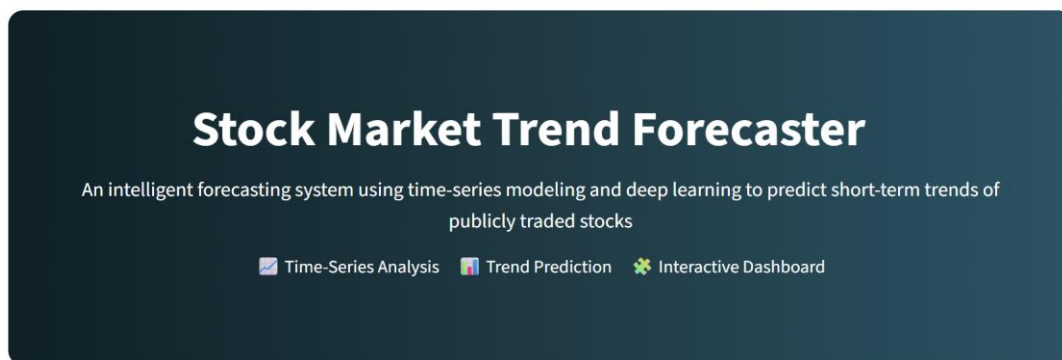
Milestone	Description	Date Achieved
Project Kickoff	Initiated the internship project	2/2/26
Prototype/First Draft	Performed preprocessing	13/2/26
Mid-Term Review	Handled missing values, outliers	23/2/26
Final Submission	Perform model training And created a interactive dashboard	6/3/26
Presentation	Completion of final report	13/3/26

5b. Project execution details

- The project was executed through a structured workflow beginning with the collection of historical Apple Inc. stock data from the Yahoo Finance API.
- The dataset was then preprocessed by removing duplicate records and handling missing values using statistical imputation techniques.

- Outliers were identified and removed using IQR and Z-score methods to improve data quality and model performance.
- Feature scaling using Min-Max normalization was applied, and time-series sequences were created using the previous 60 days of stock prices.
- Finally, prediction results were evaluated using RMSE and MAE metrics and visualized through graphs and an interactive dashboard.

6. Snapshots / Screenshots



Stock Dataset Retrieval

Stock Symbol

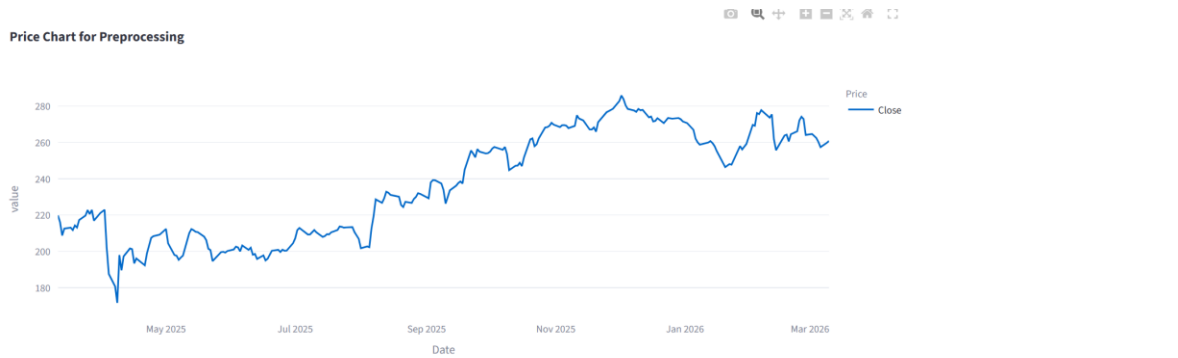
AAPL

Date Range

1y

Sample Dataset

Date	Close	High	Low	Open	Volume
2025-03-11 00:00:00	219.8817	224.86	216.5064	222.8388	76137400
2025-03-12 00:00:00	216.0384	220.7877	213.9774	219.1847	62547500
2025-03-13 00:00:00	208.7701	215.899	207.5156	215.0129	61368300
2025-03-14 00:00:00	212.5636	213.0215	208.6705	210.3333	60107600
2025-03-17 00:00:00	213.0714	214.2861	209.0589	212.3844	48073400



Model Training

Model Training Completed

Model Performance Metrics

RMSE

9.55

MAE

8.13

MAPE

3.09

Next Day Prediction Result

Today's Price: \$260.83

Predicted Next Day Price: \$265.67

Predicted Trend: UPWARD 📈

The predicted stock price is higher than today's price indicating a forward upward movement.

7. Challenges Faced

- ❖ Handling missing values and inconsistencies in the stock dataset was a major challenge and required careful preprocessing.
- ❖ Model overfitting was another concern due to the complexity of financial time-series patterns.

- ❖ Selecting suitable hyperparameters such as sequence length, epochs, and batch size required multiple experimental trials.

8. Learnings & Skills Acquired

- This internship provided valuable experience in applying machine learning techniques to real-world financial data.
- It strengthened skills in data preprocessing, including handling missing values, removing duplicates, and detecting outliers.
- The project also improved knowledge in time-series analysis using LSTM models and enhanced analytical and problem-solving skills.

9. Testimonials from team

- Working on the StockTrendAI project during the Infosys Springboard Virtual Internship was a valuable learning experience.
- The project provided practical exposure to applying machine learning and deep learning techniques for stock market prediction.

10. Conclusion

- The StockTrendAI project demonstrates how deep learning, particularly LSTM models, can be used to analyze financial time-series data and predict stock market trends.

- The system achieved reasonable prediction accuracy and presents results through an interactive dashboard, highlighting the potential of AI in financial forecasting.

11. Acknowledgements

- ❖ We sincerely thank **Infosys Springboard** for providing the opportunity to participate in the Virtual Internship and work on this project.
- ❖ We also appreciate the mentors and team members for their guidance, support, and collaboration in successfully completing the project.